

Project 2
EAE 135 Winter 2026

Deadline: Saturday February 21st, 2026, Tuesday February 24th, 2026, at 11:59 pm
on Canvas

(if working in teams, upload only one submission)

Objectives

To familiarize yourself with a simplified Euler-Bernoulli composite beam under a combined axial loading and a 1D bending load; to familiarize yourself with an interesting launch rocket system.

Scenario

Your internship at a startup founded by former students of the Cape Peninsula University of Technology (CPUT) in South Africa is going great! The Africa-European Union Space Partnership, a consortium led by the African Space Agency (AfSA, <https://africanspaceagency.org/>) and the European Space Agency (ESA, <https://www.esa.int/>), has responded to your previous report. While their technical team (your graders) reviews and validates your results, your work has sparked additional curiosity, and they are now requesting a more detailed analysis of the rocket system. Below is the response letter that your supervisors received following the LOI, outlining the requests and expectations for your next submission.

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Title, NAME
African Space Agency
Egypt Space City
New Cairo Governorate, 4764410, Egypt

February 8, 2026

Vela Cape
Cape Peninsula University of Technology
CPUT Bellville Campus, Symphony Way
Bellville South, Cape Town, 7530, South Africa

Object: Response to LOI and Request for Further Analysis

Respected [NAME],

On behalf of the Africa-European Union Space Partnership, thank you for your Letter of Intent regarding RFP [Number] and your proposed launch solution comprising the BlueSky-B1 booster and Horizon-L1 launcher for deployment of a CubeSat constellation.

Our technical review team has begun validating the results and assumptions presented in your preliminary report. While this validation is ongoing, your work has generated interest and has prompted several follow-up questions regarding vehicle performance and design margins. In order to support a thorough evaluation of feasibility and risk, we request a more detailed analysis addressing the items listed below.

We appreciate the alignment of your mission objectives with the Partnership's goal of expanding access to orbit and strengthening technical capacity for Earth observation missions across Africa. We look forward to reviewing your updated analysis and continuing the evaluation process.

Respectfully,

Signed and dated

[NAME]

[Title]

For the Africa-European Union Space Partnership

[END OF LETTER]

You have been asked to make a very rough estimate of the structural integrity of the first stage of the Horizon-L1 rocket, at a set time of its flight, using the Euler-Bernoulli beam theory taught in EAE 135. It is understood that you will need to make many simplifying assumptions for this task.

The Project

In Project 1, you have worked at $T + 5$ seconds, where t is the time of the drop, corresponding to when the rocket stops its free fall and is ignited still in the horizontal position. The maximum thrust of stage 1 of 726 kN. The rocket diameter is approximated at 128 cm with the propellant diameter of 118 cm. Assume that dynamic effects and gravity are neglected, and that this is not a buckling analysis. For Project 2, Horizon-L1 is now turning upward with the assistance of its Delta wing and fins, and reaches its **Max Q** (maximum aerodynamic pressure) at $T+36$ seconds. Max Q is the moment of the rocket's highest mechanical stress and is a key milestone of a rocket launch.

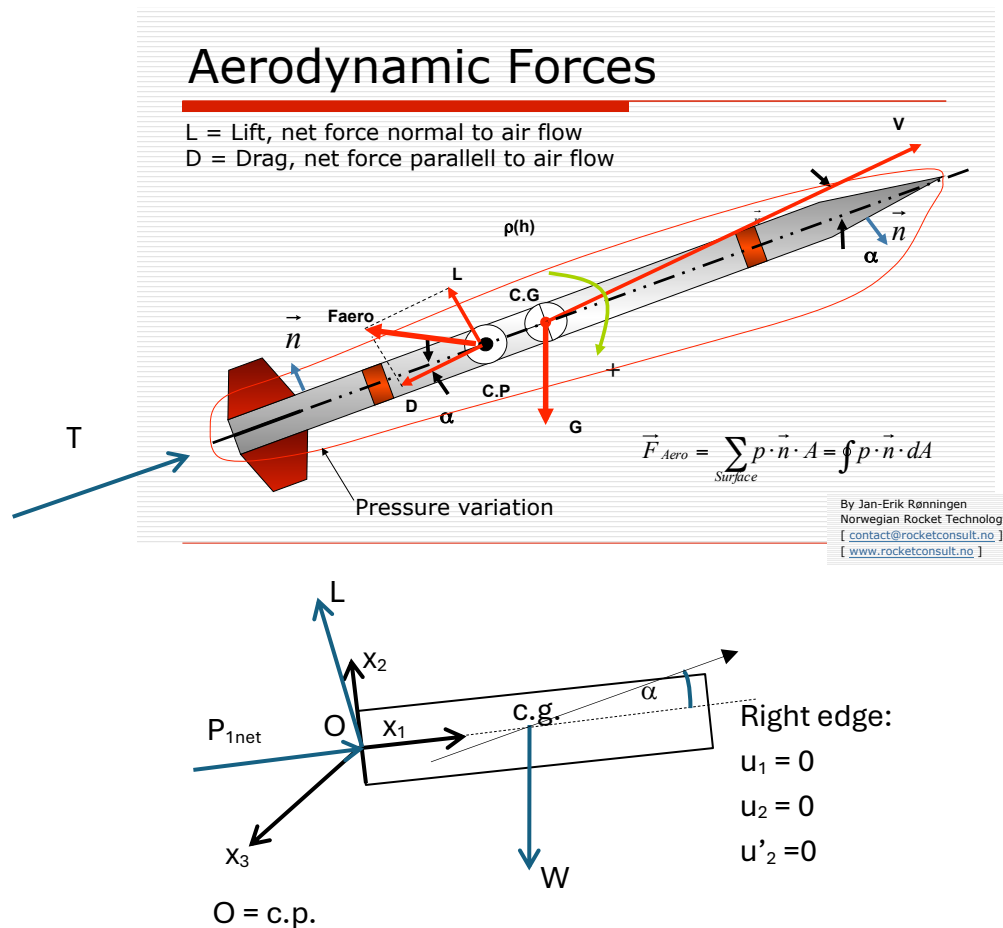


Figure 1. (Top) forces in a rocket.
 (Bottom) simplified model for a section ℓ of the first stage (not to scale).

Assume for simplicity that:

- 1) You are studying only a section of length ℓ smaller than the actual length of the first stage.
- 2) The center of pressure c.p. and the modulus-weighted centroid C of that cross-section coincide with the origin O of the reference system.
- 3) The center of pressure equal to O is located to the left end side of the section of interest.
- 4) The center of gravity c.g. is located one caliber away (caliber = one diameter of the rocket), as required for rocket stability. The modulus-weighted centroid of every cross-section is on the axis of the rocket.
- 5) The boundary conditions at the left end side of the section of interest are two concentrated forces located at O: P_{1net} , the resultant of thrust and the projection of the drag along the axis of the beam, and the lift L. On the right end side, assume no axial and transverse displacements and no slope.
- 6) The rocket casing to be made of black aluminum $[0/\pm 45/90]_s$ with AS4 carbon/epoxy layers (note: we do not know the actual composition of the casing, and we will use this layup because of its common use in aerospace structures).
- 7) The propellant is HTPB, aged 185 days, with the highest stiffness and strength for this material. Additional information is given in **Table 1**.

Table 1. Input data.

Property	Value
Angle of attack α	20 degrees
Beam length ℓ	5 diameters
Thickness of one layer	1/8 of 50 mm
Distance c.p.-c.g.	1 diameter of the first stage
Net axial force on O, P_{1net}	600 kN (estimated)
Weight W	8500 kg (estimated) –ignore axial component
Material properties for 1 layer	See AS4/epoxy, p. 3 of tables_composites.pdf
Strengths for 0 deg. layers	Axial strength values, p. 3 of of tables_composites.pdf
Strengths for ± 45 layers	Transverse strength values, p. 3 of of tables_composites.pdf
Strengths for 90 layers	Transverse strength values, p. 3 of of tables_composites.pdf
Lift	1400 kN (estimated)
Safety factor	1.25

Miscellaneous Information

You can work in teams up to 3 people. The submitted report should be professional and easily readable. Typing is best, but neat handwriting is acceptable. We are asking for the results to be in tables and plots (see below).

The recorded presentation format is up to you (e.g. use of a board, PowerPoint, tablet, single presenter or group of presenters), but we do expect professionalism. You are not required to dress up but we expect you not to be in a swimming attire or in a PJ.

AI Policy

Students are not allowed to use advanced automated tools (artificial intelligence or machine learning tools such as ChatGPT) on any assignment of this course.

Resources for this Project

- Paper by N. Gligorijevic et al., “Mechanical Properties of HTPB Composite Propellants in the Initial Period of Service Life”, on Canvas
- Third table in the document tables_composite.pdf, for AS4/epoxy. Convert into SI units.
- Calculation_bendingstiffness_laminate_EAE135.pdf for the analysis of a laminated cylinder
- Bauchau & Craig textbook, sections of Chapter 5: particularly, 5.5.4, 5.5.5, 5.5.6, 5.6.4, Example 5.8 (cantilever beam with concentrated load).

Deliverables

You are asked to submit **three items**.

Item 1: a professional report answering the following questions:

1. Calculate the solutions u_1 and u_2 of the two governing equations related to this problem.
2. Find the location $x_{1\text{critical}}$ with the highest axial stress (in absolute value); at that cross-section, **plot the axial stress distribution** (in the x_1 axis of the plot) along the thickness (in the x_2 axis of the plot). Zoom in the laminated casing if needed, to show details (i.e. change your axis limits).
3. Check whether the casing fails anywhere for x_1 between 0 and ℓ . Present the maximum axial stress results and the failure check (yes/no failure) **in a table**.

Comment on the steps and results.

For the analysis, ignore any dynamic effects, buckling, temperature effects, any bending along axis x_2 , torsion due to pitching. Ignore the presence of the other two stages plus payload.

Item 2: a recorded presentation. Include the link of the video inside the report. Check before submission to make sure the video is viewable.

The video time length is constrained to **5 minutes**. The face of the presenter(s) need(s) to be visible during most of the video.

The presentation needs to address the three aspects below.

- 1) Problem statement: what is the problem to solve
- 2) Solution method: what methods were used, and what are their hypotheses
- 3) Discussion of the results: how results are used, and do they make sense

The presentation will be graded also for 1) clarity and professionalism, 2) the ability to convince the instructor that the student(s) understand(s) the topic and did the work, 3) recording quality.

Instructions for uploading the video of the recorded presentation

- Go to video.ucdavis.edu
- After entering your login credentials (top right of your screen), click on top right corner, '+Add new'.
- Drag the video you recorded to the 'Upload media' field.
- Click on **Publishing status** → **select 'unlisted'**
- Enter description: your name, recorded presentation number #
- **Save**

After the video has finished uploading and you see green-fielded 'your changes have been saved', go back to your name on the top right corner, and click on My Media. You will see the video being processed/compressed. Click on the video. You will see **Details Share Symbol of a magnifying glass**. Click on 'Share'. You will see a link. **Copy and paste the link into your submission.**

The video must be checked prior to submission. The link must be checked to make sure it is working properly and can be viewed. In the past, people forgot to follow instructions.

There will be a 5% penalty if the video is not viewable and we need to contact you to upload it properly. Please do not erase the video until the quarter is over, in case we decide to go back and review it. Please **DO NOT** add us as collaborator/co-viewer.

Item 3: an appendix with references used, workload distribution (in a team of up to 3), any code used (in text form and readable). If you used a spreadsheet, upload it as a separate file.

If you worked individually for Project 1 and decide to team up for Project 2, you are required to email the instructor (vlasaponara@ucdavis.edu email only) **no later than Tuesday 02/17 at 2:30 pm.**

Use the video recorded on Thursday 02/12 giving an overview of the project, and the accompanying document "use_for_project2.pdf", to get started on the project.

Grading Rubric

EAE 135 Project 2, Winter 2026

Grading Rubric (100 point scale)

Authors: _____ || _____

Technical Requirements of Report

_____/40

- Equations with units where applicable _____/10
- Correct calculations of u_1 and u_2 _____/20
- Correct calculation of axial stresses, correct failure analysis _____/10

Delivery of Report

_____/30

- Clear rhetorical structure (thoughts and concepts are presented in an organized, logical structure) _____/10
- Quality of text (neat, professional) _____/10
- Calculations are explained _____/10

Recorded Presentation

_____/30

- The video presentation does not exceed 5 min. time limit; the presentation is professionally delivered, with clear audio, eye contact. The face of the presenter(s) appear(s) often. _____/10
- The presentation is clear and concise and answers all aspect (problem statement, solution, results discussion) _____/20

TOTAL

_____/100